

Foto: Dr. Matthias Otte

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EUROfusion

HELMHOLTZ

RESEARCH FOR GRAND CHALLENGES



IMPURITY TRANSPORT AND RADIATION AT THE STELLARATOR WENDELSTEIN 7-X

Promotionskolloquium

March 26, 2026

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CONTENTS

MOTIVATION

NUCLEAR FUSION: ENERGY CHALLENGE

Lawson criterion for net energy gain:

$$n_e T \tau_E \geq \frac{12 f_{\text{tot}}}{\langle \sigma_{\text{DT}} v \rangle f_H^2 E_\alpha - 4 L_Z(T)} T^2$$

$$\gtrsim 3 \times 10^{21} \text{ keVs/m}^3$$

(requires temperatures $T \sim 10^1$ - 10^3 keV)

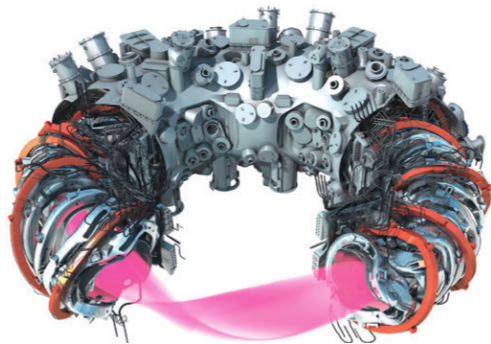
Triple Product

Key figure of merit for fusion performance combining density, temperature, and energy confinement time

WENDELSTEIN 7-X

Key Challenge

Managing heat loads on plasma-facing components during long, high performance plasma



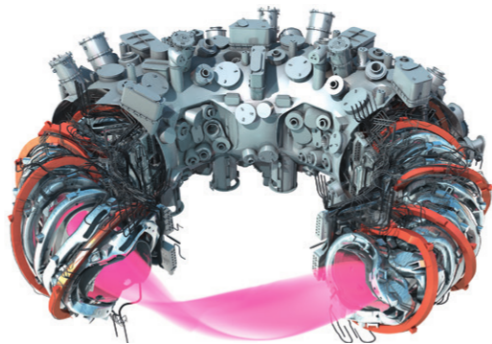
Source: © IPP

WENDELSTEIN 7-X

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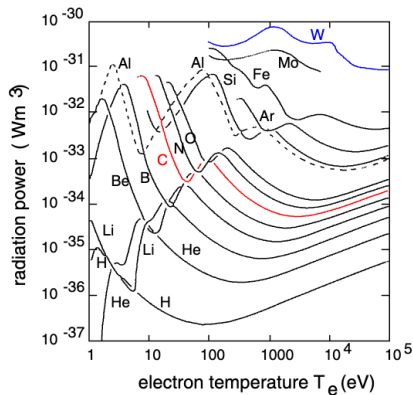
⇒ Dissipation of large fraction of heat load (energy) in scrape-off layer and particle flux to the target through detachment



Source: © IPP

PLASMA RADIATION

- adjust plasma profiles so radiation in SOL increases and transport across seperatrix reduced



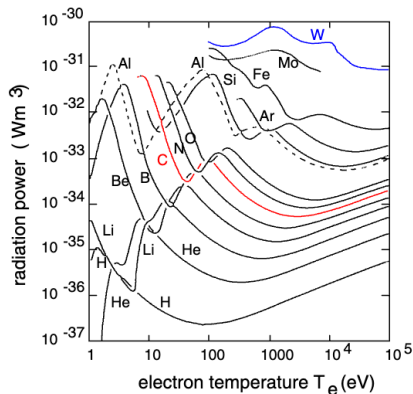
Source: R. Schneider et al., 2006 [Schneider2006]
 $P_{\text{rad}} / (V n_e n_{\text{imp}})$ as a function of electron temperature

PLASMA RADIATION

- adjust plasma profiles so radiation in SOL increases and transport across separatrix reduced

Impurity Seeding

Injection of low-Z (He, N₂) or high-Z (Ne, Ar) impurities to enhance edge radiation and cooling



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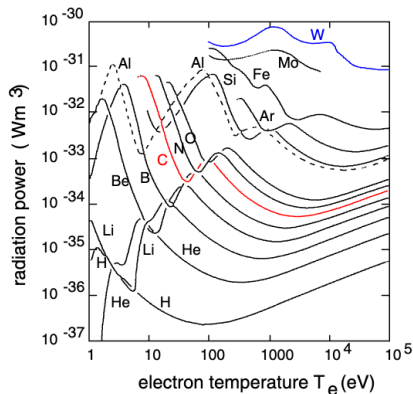
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Impurity Seeding

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- avoid critical impurity accumulation, use working gas

⇒ stable control w. radiation information?



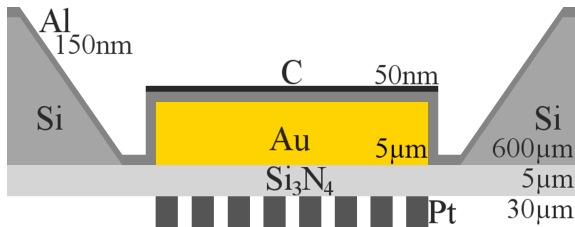
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CORE BOLOMETER AT W7-X

BOLOMETER DIAGNOSTIC

Metal Resistor Bolometers:

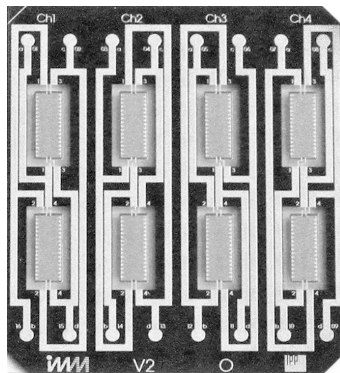
- Thin metal (gold) film absorbers
- Wheatstone bridge circuit configuration
- Multicamera system: HBC & VBCI/r (32 + 2×24 channels)
- Spatial resolution: ~ 5 cm at magnetic axis



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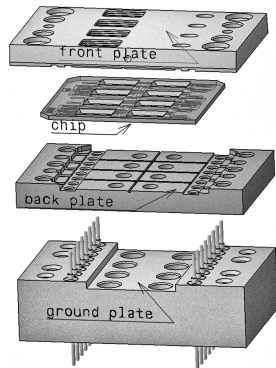


Source: D. Zhang et al., 2010[Zhang2010]

BOLOMETER DIAGNOSTIC

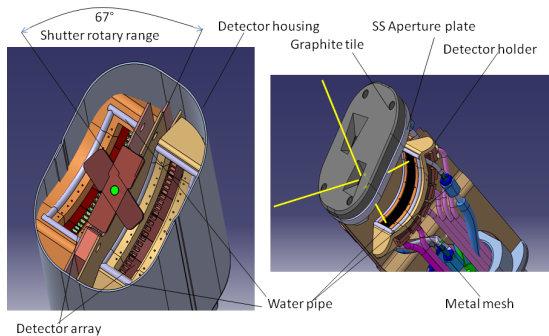
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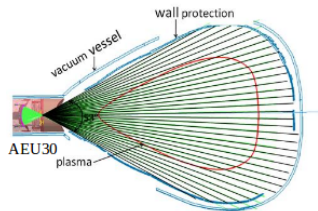


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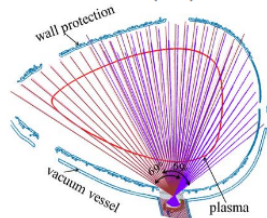
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Source: D. Zhang et al., 2010[Zhang2010]



horizontal bolometer camera (HBC)

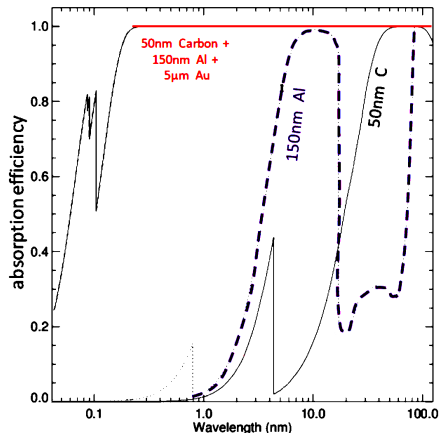


AEU21
vertical bolometer camera (VBC)

PERFORMANCE

Performance Characteristics:

- Measures total radiation power (broadband)
- Near unity absorption efficiency 600-0.2 nm
- In-situ ohmic heating calibration with Wheatstone bridge (F_M , f_M)
- $\{0.4, 0.8, \dots 6.4\}$ ms sampling
- *Gaussian* uncertainty:
 $1.6 \mu\text{W} \leftrightarrow \text{SNR} > 1000$



Source: D. Zhang et al., ~2024[Zhang2024]
Relative absorption efficiency of absorber assembly

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Power Per Detector:

$$P_M = F_M \cdot \left(\frac{d(\Delta\tilde{U}_M)}{dt} + f_M \Delta\tilde{U}_M \right)$$

Global Radiation Power:

$$P_{\text{rad}} = \frac{V_{\text{P,tor}}}{V_C} \sum_M \frac{P_M V_M}{K_M}$$

PERFORMANCE

Chord Brightness

Line-integrated measurement along each detector's line of sight provides radial profile information ($P_{\text{ch}} \sim P_M/K_M$)

Applications:

- Power balance calculations
- **Tomographic reconstruction**
- **Real-time feedback control**

⇒ controlled cooling in the plasma edge

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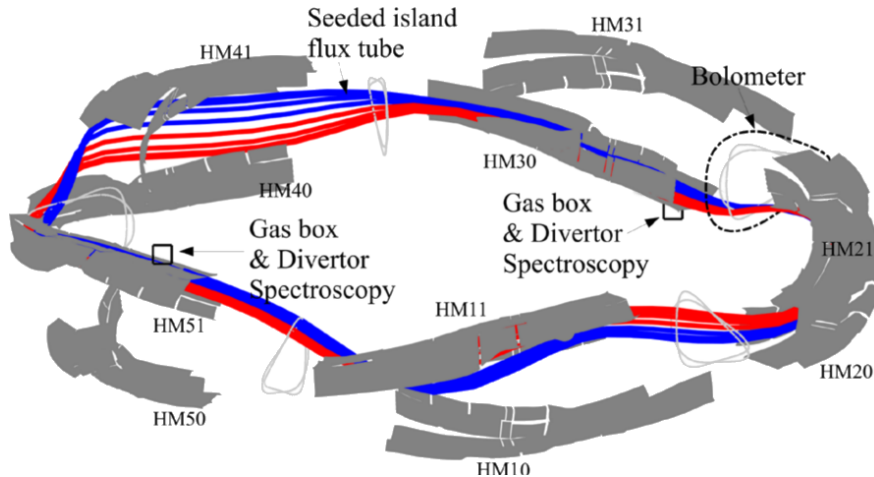
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REAL-TIME RADIATION FEED- BACK CONTROL

FEEDBACK SYSTEM



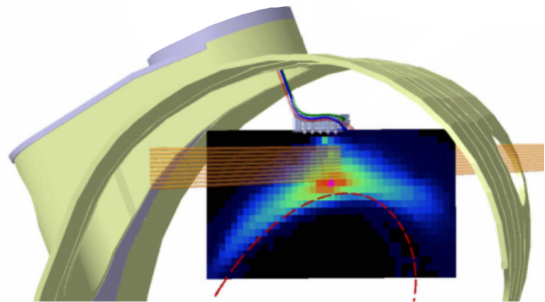
Source: F. Effenber et al., 2019[Effenberg2019_seed]

Divertor elements HM 1-5: up (1) down (0) & gas boxes, bolometry system and connecting field lines

FEEDBACK SYSTEM

Architecture:

- Integrated data acquisition hardware & optimized control software
- Twin proxies and dedicated signal outputs
- Thermal gas valve actuation with multi-diagnostic PID controller



Source: T. Barbui et al., 2016[Barbui2016]
Thermal helium beam and spectroscopy

FEEDBACK SYSTEM

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Radiation Prediction Proxies:

A: *Channel selection* S

$$P_{\text{pred}}^{(1)} = \frac{V_{\text{P,tor}}}{V_S} \sum_M^S \frac{P_M V_M}{K_M}$$

(e.g. $S_{\text{VBC}} = \{61, 86, 78, 73, 70, 65\}$)

B: *Single channel*

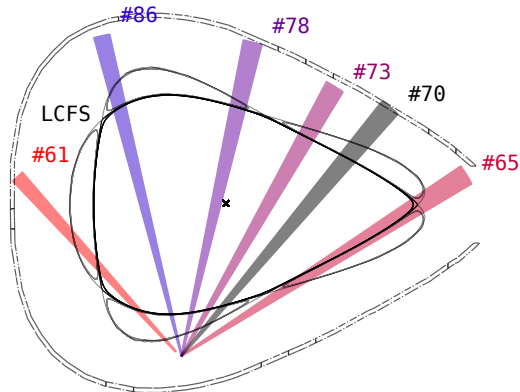
$$P_{\text{pred}}^{(2)} \propto \Delta \widetilde{U}_M$$

(dimensionless, minimal overhead)

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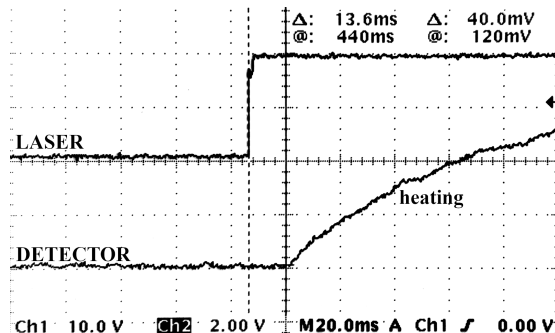


Line of sight selection S (XP20181010.32), only VBC channels covering magnetic islands, as well as the core and X-points partially

FEEDBACK SYSTEM

Performance:

- FIFO smoothing in $P_{\text{pred}}^{(1)}$: $\sim N\Delta t/2$
- Minimum acquisition latency 13.6 ms
- Total system latency: 150-400 ms (dominated by plasma response)



oscilloscope screen capture; single detector
 laboratory latency measurement with laser diode

EXPERIMENTAL ACHIEVEMENTS

XP20181010.32

- Radiation fraction PID setpoint

$$f_{\text{rad}} \sim 90\% \text{ off } P_{\text{pred}}^{(1)}$$

- 5-channel VBC subset

$$S = \{61, 65, 73, 78, 86\}$$

EXPERIMENTAL ACHIEVEMENTS

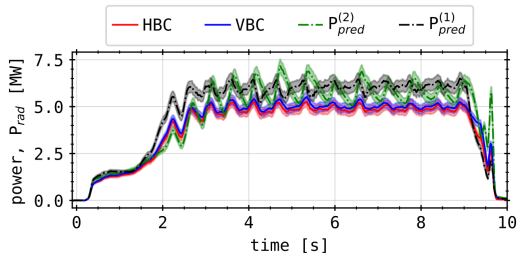
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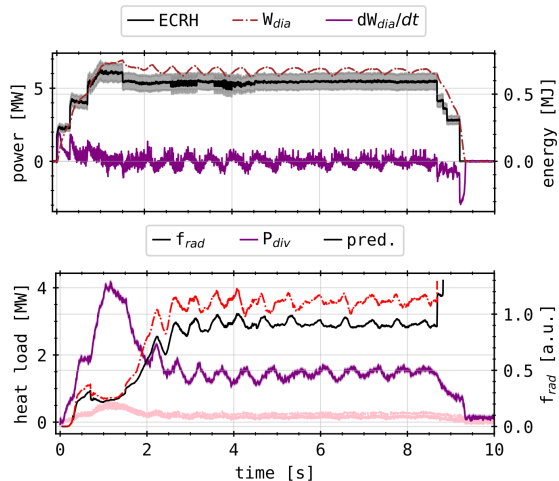
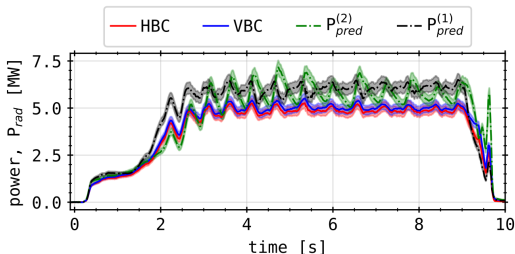
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EXPERIMENTAL ACHIEVEMENTS

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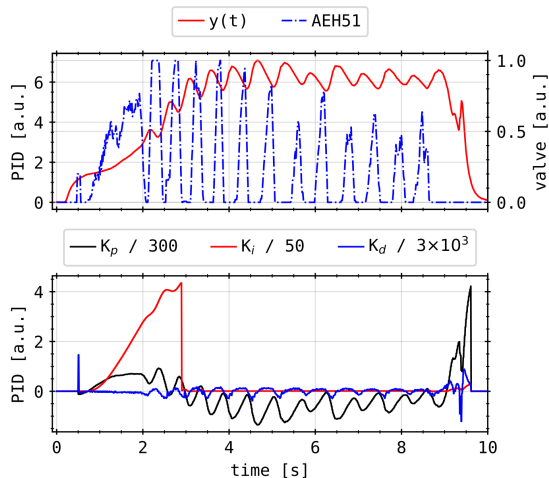
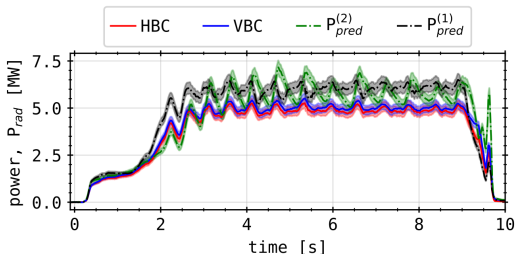
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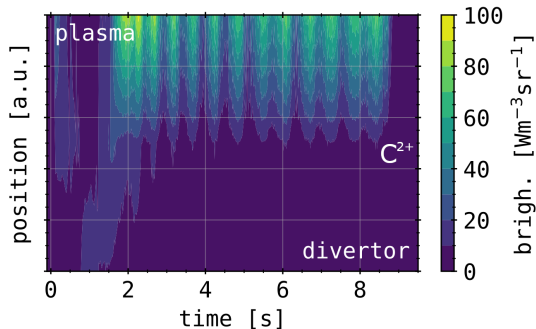
EXPERIMENTAL ACHIEVEMENTS

XP20181010.32

- Peak $f_{\text{rad}} \gtrsim 100\%$, cycling with gas inlet
- Target heat load reduction factor ≥ 2
- C^{2+} detachment visible at $f_{\text{rad}} \sim 50\%$

Key Result

Demonstrated stable, real-time feedback-controlled radiative cooling with $f_{\text{rad}} \geq 85\%$ without terminal plasma disruption and significant target heat load reduction



spatial-temporal evolution (C^{2+} radiation)
in divertor of HM51; filterscope measurements

LINE-OF-SIGHT SENSITIVITY

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- Evaluate LOS/channel efficacy for feedback (generalised local radiation sensitivity)
- Data set of all feedback discharges from OP1.2b
- Applied 5 different metric (linear & weighted deviation, FFT, ...)

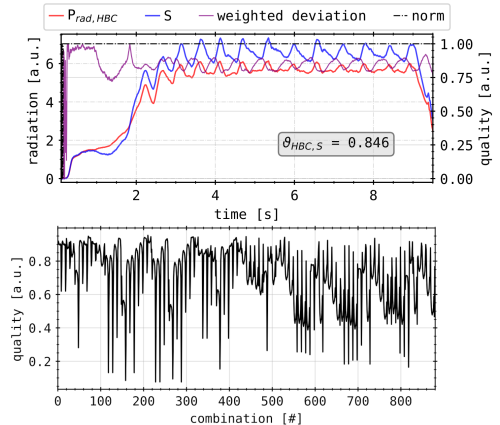
Weighted deviation:

$$\varphi(t) = 1 - \frac{\|P_{\text{pred}} - P_{\text{rad}}\|(t)}{P_{\text{rad}}(t)}$$

$$\vartheta = \frac{1}{T} \int_T \varphi(t) dt$$

LINE-OF-SIGHT SENSITIVITY

- Evaluate LOS/channel efficacy for feedback (generalised local radiation sensitivity)
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(top) weighted deviation norm and example for $S_{HBC} = \{8, 7, 23\}$;
 (bottom) full dataset for $M = 3$ selection sets (HBC)

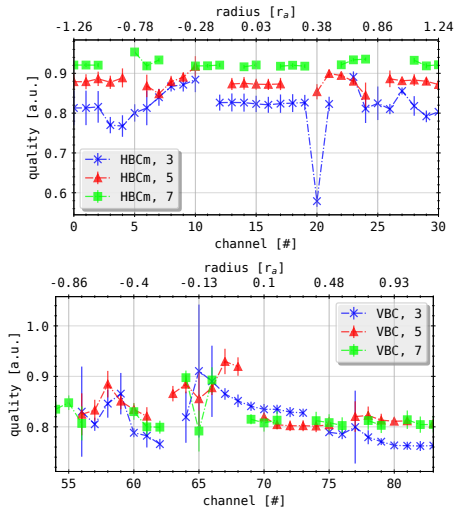
LINE-OF-SIGHT SENSITIVITY

Optimal Channel Selection:

- No single "best" set exists
- Robust selection achieves $\geq 85\%$ prediction accuracy
- VBC channels generally outperform HBC for edge sensitivity
- 3-5 channels sufficient for reliable P_{pred}

Takeaway

Detectors viewing close to separatrix and SOL most viable for feedback purposes



LOS sensitivity by weighted deviation measurement for $M = 3, 5, 7$

TOMOGRAPHY

MINIMUM FISHER REGULARIZATION

Measurement:

$$\mathbf{T} \vec{x} = \vec{b}$$

Bolometer tomography:

- 61 line-integrated measurements $\vec{b}$
- 2D emissivity $\vec{x}$: ~ 4500 pixels
- *Geometry* $\mathbf{T}$ has high condition

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Bolometer tomography:

- 61 line-integrated measurements $\vec{b}$
- 2D emissivity $\vec{x}$: ~ 4500 pixels
- *Geometry* $\mathbf{T}$ has high condition
- **ill-posed, requires regularization**

Problem:

$$\min_{\eta} \left\| \left(\mathbf{T}^T \mathbf{T} + \mu^{-1} \mathbf{H} \right)^{-1} \mathbf{T}^T \vec{b} \right\| \rightarrow \vec{x}$$

Fisher Information:

$$I_F = \int \frac{1}{x} \left(\frac{\partial x}{\partial \vec{r}} \right)^2 d\vec{r} \leftrightarrow \mathbf{H} \sim \frac{1}{x} \Delta^T \Delta$$

limiting error on reconstruction:

$$\sigma_x \leq \frac{1}{I_F}$$

MINIMUM FISHER REGULARIZATION

Radially Dependent Anisotropy:

- Tailored weighting to poloidal gradient
- $k_{\text{ani}} < 1$: favours anisotropy
- $k_{\text{ani}} > 1$: favours smooth emissivity

$$\mathbf{H} \rightarrow \mathbf{H}_{\text{ani}}$$

$$\Delta = \nabla_{\mathbf{r}}^T(\cdot) \nabla_{\mathbf{r}} + \nabla_{\vartheta}^T(\cdot) \nabla_{\vartheta}$$

$$\nabla_{\vartheta} \rightarrow \tilde{\nabla}_{\vartheta} = \frac{1}{k_{\text{ani}} \Delta r} \mathbf{D}_{\text{dia}}^{(0)} \nabla_{\vartheta}$$

MFR + RDA Advantages

- Robust to noisy data
- Incorporates a priori knowledge
- Stable convergence

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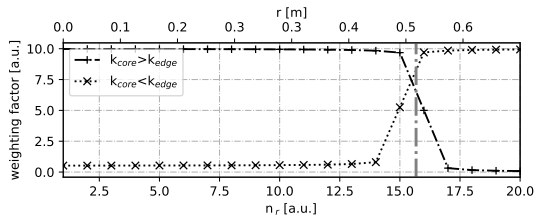
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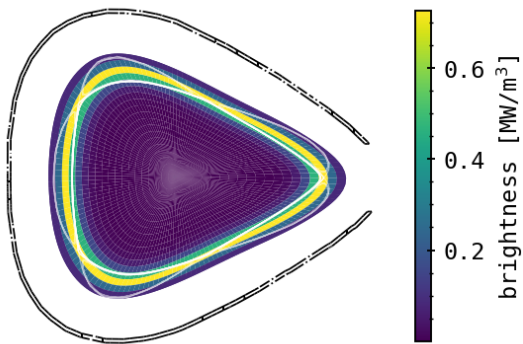
k_{ani} profile examples over radial cell number

BENCHMARKING

Geometry Perturbation Test:

- Previous and future calculations based on as-designed geometry

BENCHMARKING



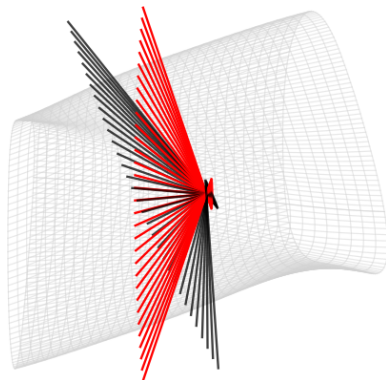
EIM grid, STRAHL phantom radiation profile
($f_{\text{rad}} \sim 100\%$)

BENCHMARKING

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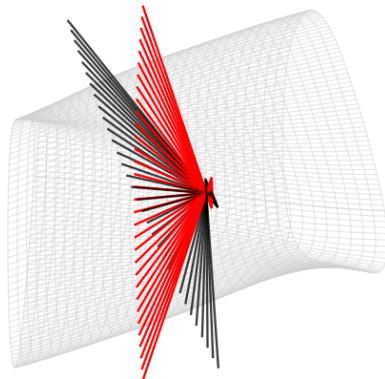
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- Forward calculation of synthetic 2D emissivity show asymmetry (STRAHL)
- Tests for symmetrical chord profile yield artificial camera geometry

BENCHMARKING



LOS comparison: original vs. artificial, symmetric HBC

BENCHMARKING



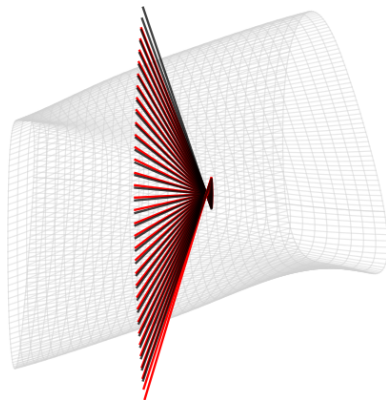
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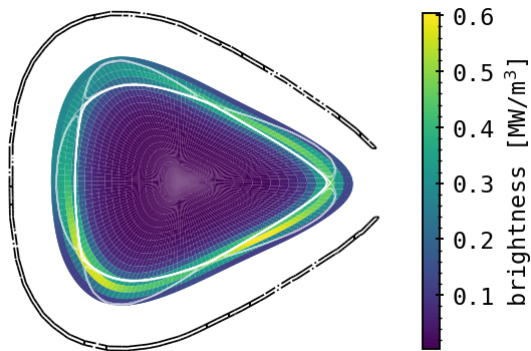
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BENCHMARKING



tilted (symmetrical) camera geometry; 2° poloidal rotation at aperture

BENCHMARKING



tomogram of forward calculation using tilted geometry, backward symmetrical
using $\mathbf{T}_{\text{sym}} \cdot \vec{x} = \vec{b}$

BENCHMARKING

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Regularized, parametric inversions need experienced tailoring for their respective application.

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Metrics:

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$$M_{\text{SD}} \sim \left\| \frac{\vec{x}_{\text{tom}} - \vec{x}_{\text{phan}}}{\|\vec{x}_{\text{phan}}\|} \cdot \vec{A}^T \right\|$$

$$P_{\text{rad},X}^{2\text{D}} = \vec{x}_X^T \cdot \vec{A}, \quad X = \{\text{phan}, \text{tom}\}$$

$$\chi^2 = \frac{1}{N_{\text{ch}}} \left\| (\vec{b}_{\text{phan}} - \vec{b}_{\text{tom}}) \vec{\sigma}_{\text{bol}}^{-1} \right\|$$

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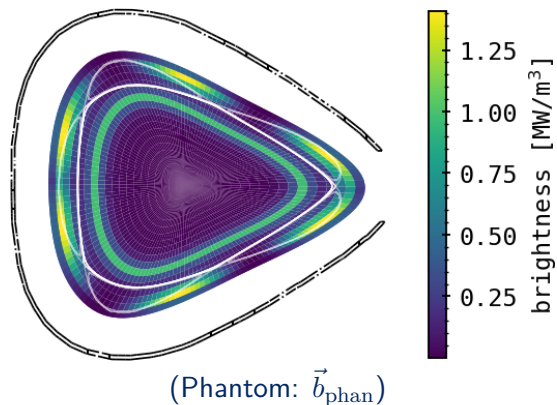
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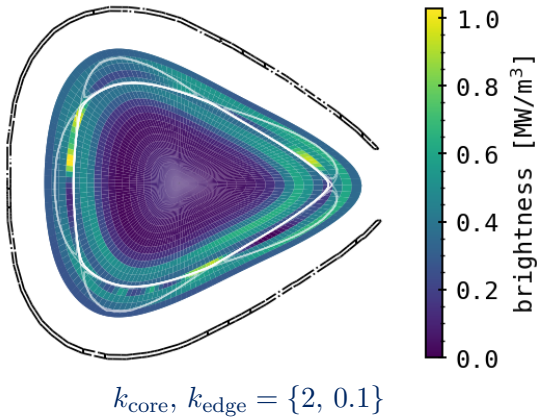
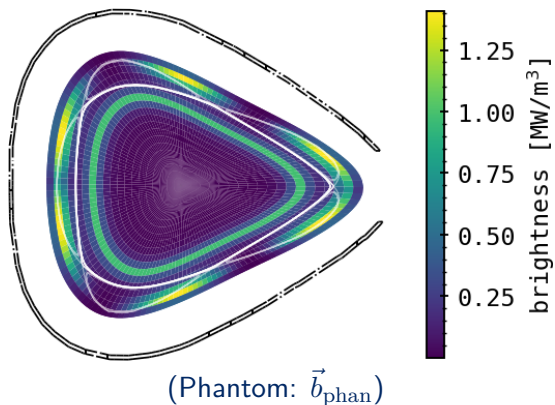
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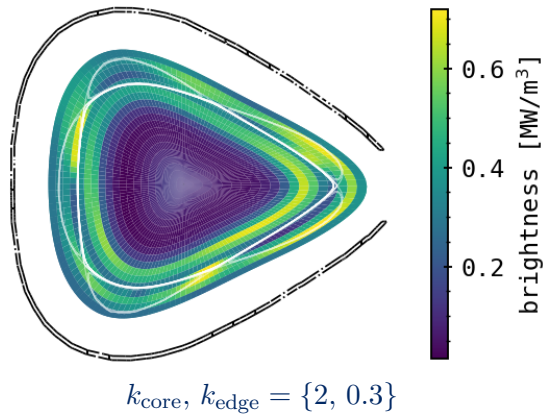
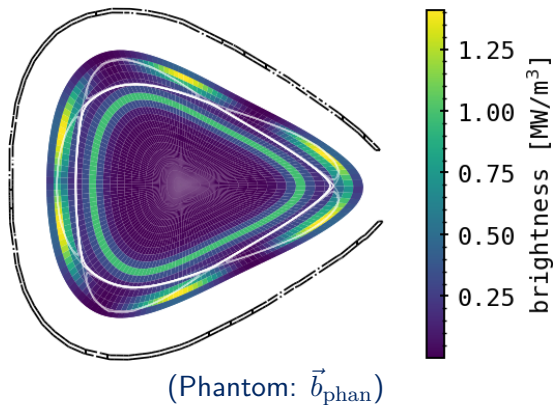
PHANTOMS: EXAMPLE



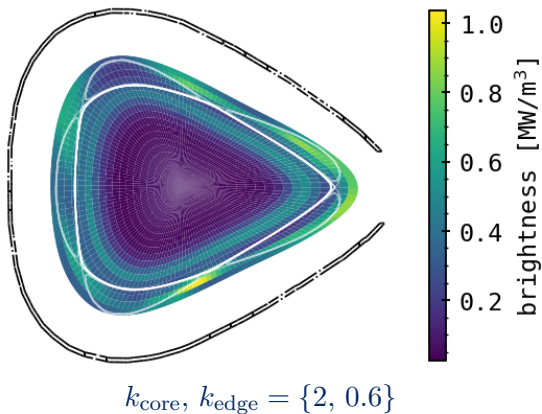
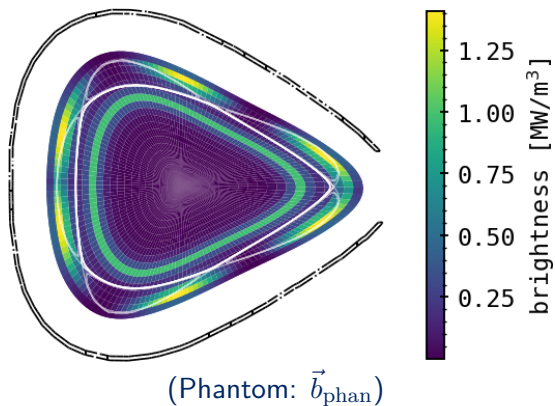
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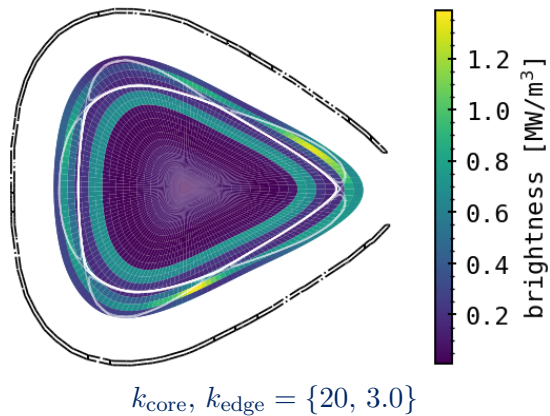
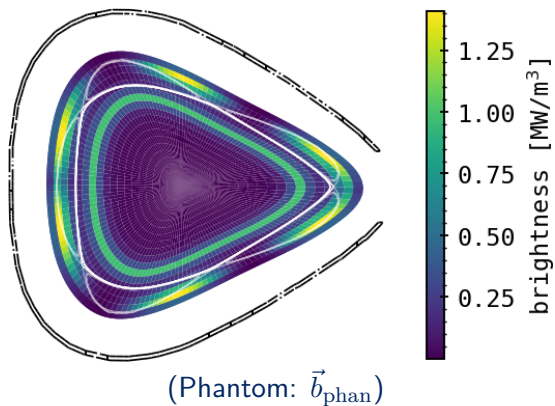
PHANTOMS: EXAMPLE



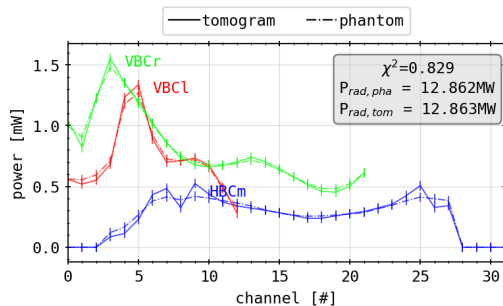
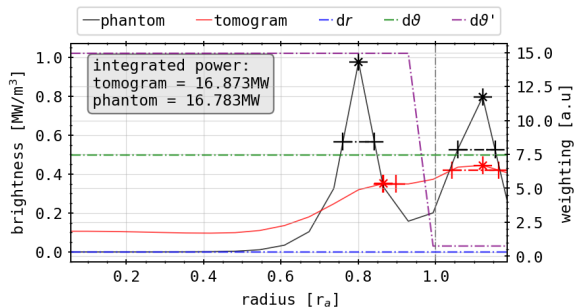
PHANTOMS: EXAMPLE



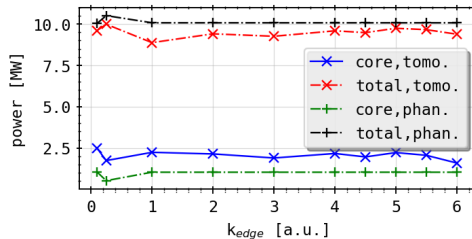
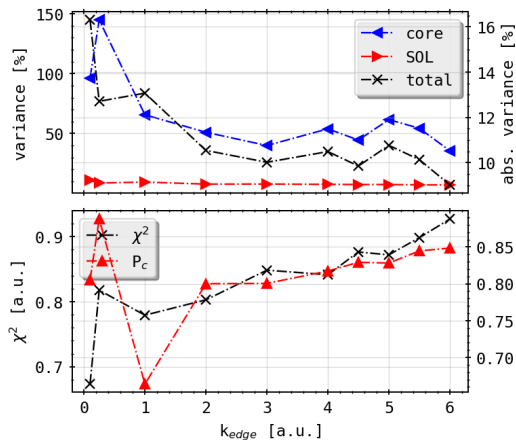
PHANTOMS: EXAMPLE



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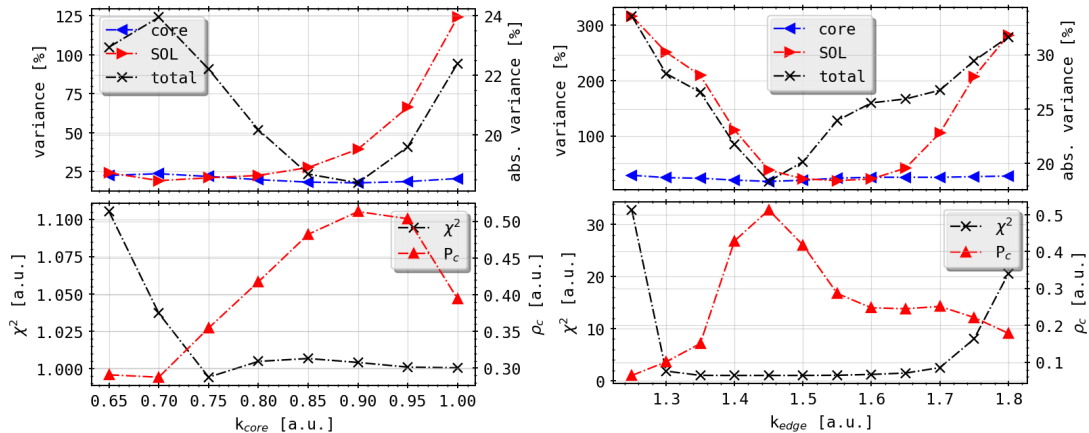


PHANTOMS: EXAMPLE



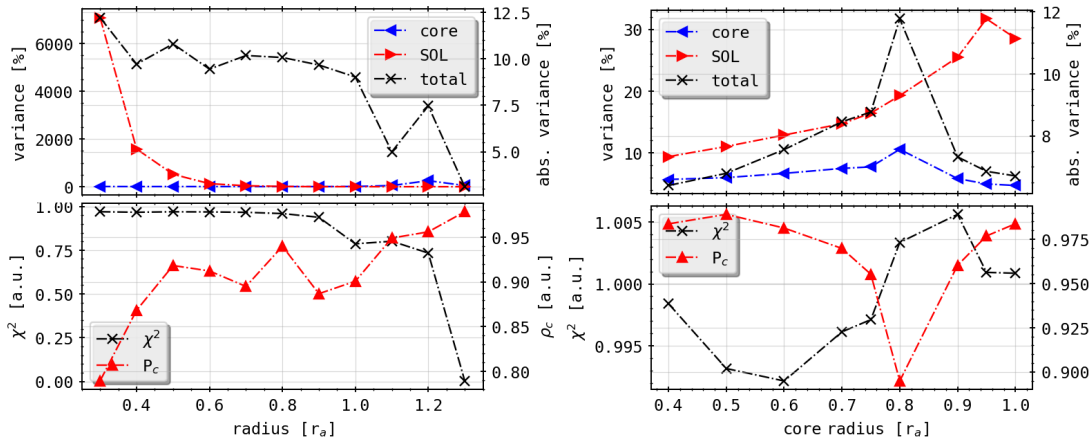
PHANTOMS: STATISTICS

Isotropic Rings: k_{ani} scan



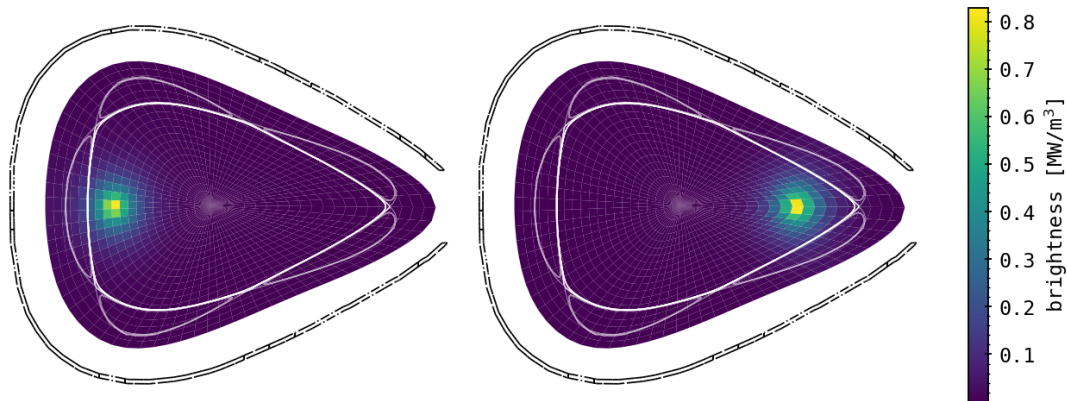
PHANTOMS: STATISTICS

Isotropic Rings: Radial Scan



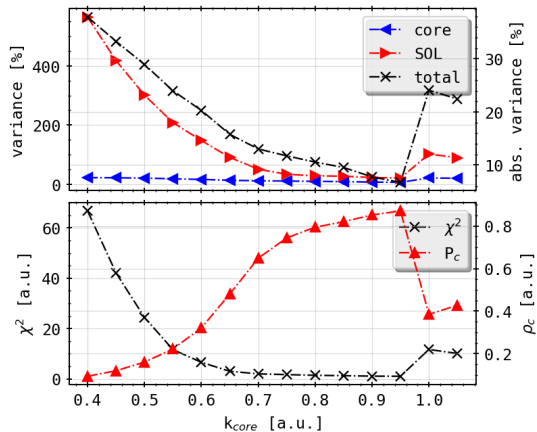
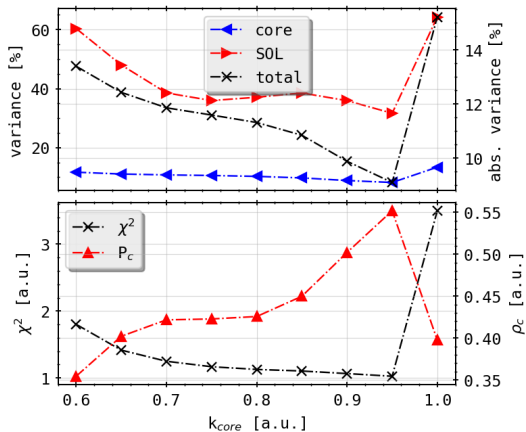
PHANTOMS: STATISTICS

Composite Distributions



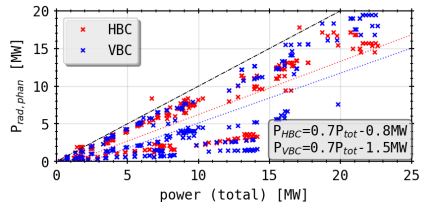
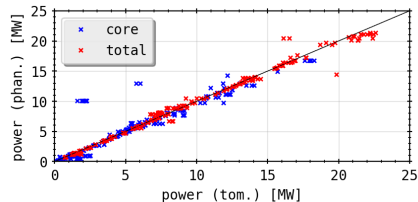
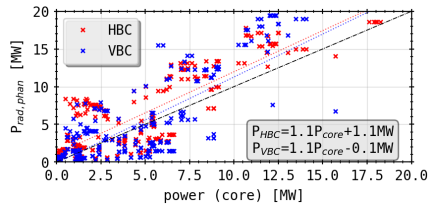
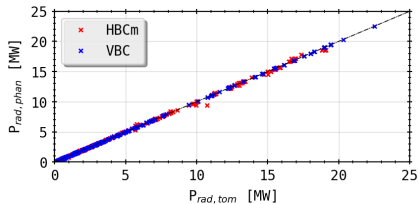
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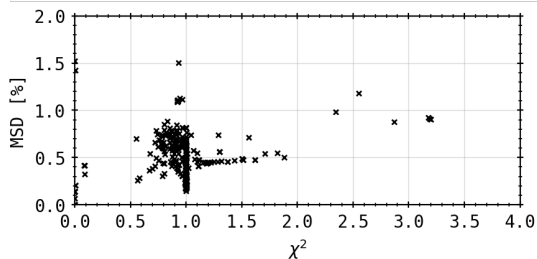
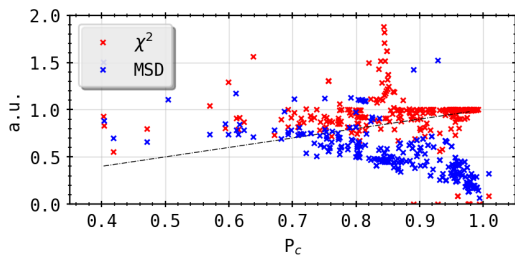
PHANTOMS: STATISTICS

Full Set



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MFR + RDA

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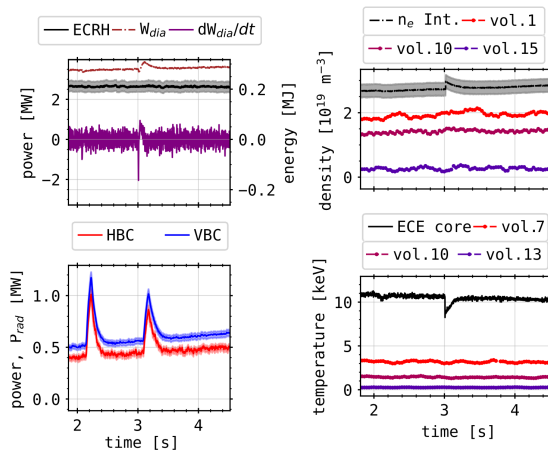
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Conclusion

MFR with RDA weighting provides robust, reliable 2D radiation reconstructions suitable for further analysis and calculations.

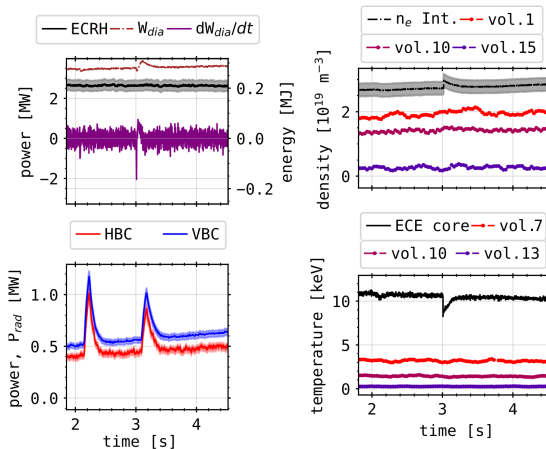
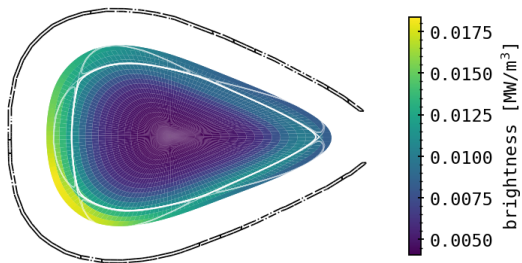
EXPERIMENTAL RECONSTRUCTIONS

Example: XP20180809.13



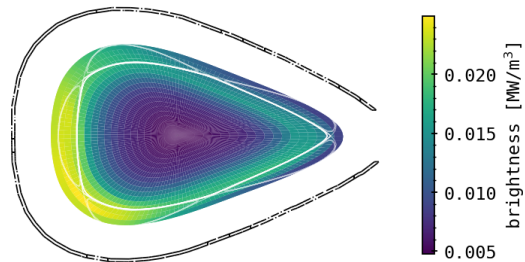
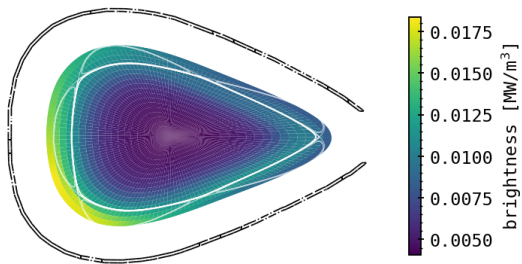
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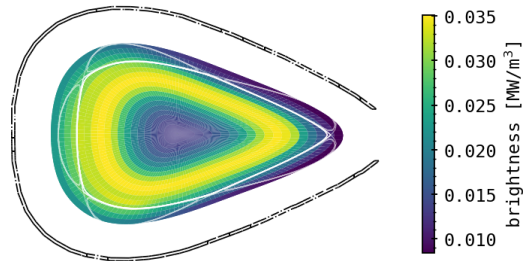
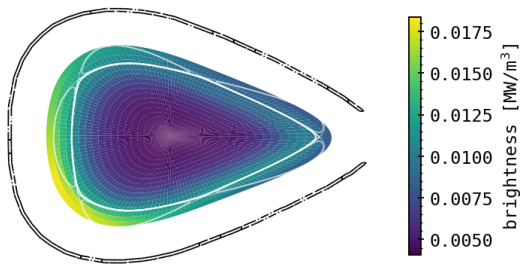
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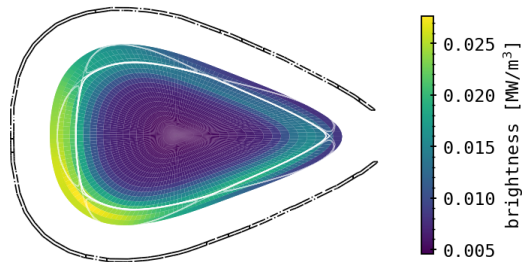
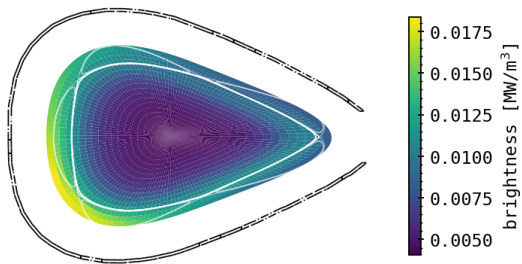
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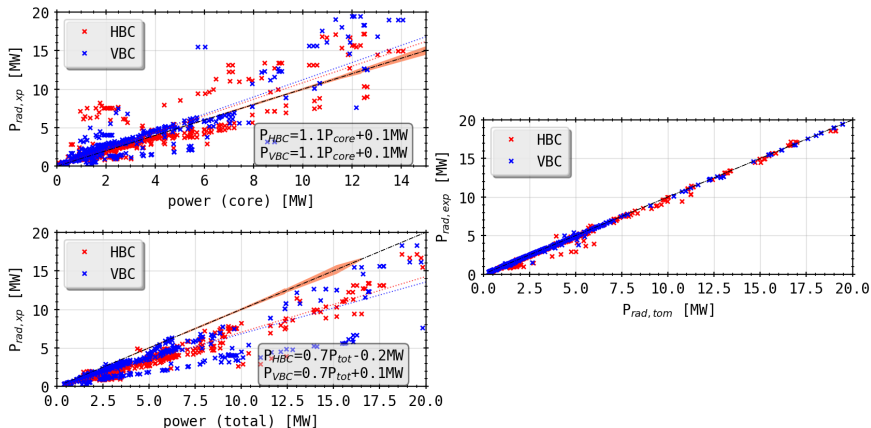
EXPERIMENTAL RECONSTRUCTIONS

Example: XP20180809.13



EXPERIMENTAL RECONSTRUCTIONS

Statistics: Full Set



GLOBAL POWER BALANCE

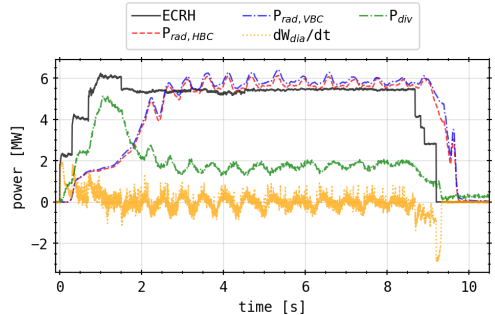
Simplified 0D Ansatz:

$$\begin{aligned} 0 &\stackrel{!}{=} S + \frac{1}{V_P} \int \frac{3}{2} \left(\frac{\partial p}{\partial t} \right) + p \nabla \vec{v} + \nabla \vec{q} d\vec{r} \\ &= -P_H + P_{\text{rad}} + P_{\text{div}} + \frac{dW}{dt} \\ &= P_{\text{bal}} \end{aligned}$$

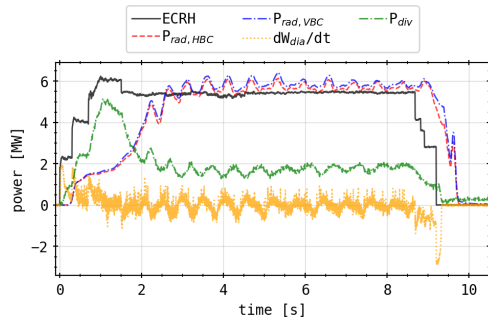
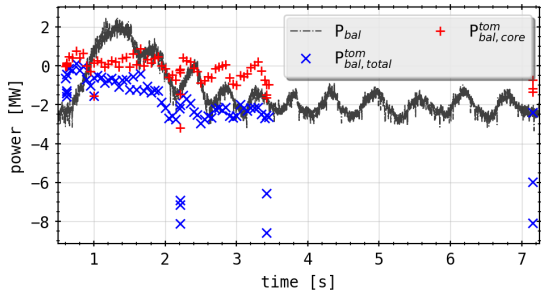
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GLOBAL POWER BALANCE



EXPERIMENTAL RECONSTRUCTIONS

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- 2D integrated total power significantly underestimated (conflict experimental experience)
- SOL emissivity over-emphasized in reconstruction and absolute power figures
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Power Balance

- $P_{2\text{D}}$ improves balance accuracy and stability
- Strong improvement over P_{rad} in transient scenarios
- Integrated 2D core emissivity nearly satisfies $P_{\text{bal}} = 0$
- Viable alternative for quasi steady-state analysis

CONCLUSIONS

KEY ACHIEVEMENTS

1. **Real-Time Radiation Feedback System:**

Implemented, tested & operated first bolometer radiation feedback at W7-X

2. **Line-of-Sight Optimization:**

Systematic sensitivity analysis for optimal detector configurations

3. **Tomographic Reconstruction:**

Benchmarked and tailored MFR algorithm with *RDA* weighting

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1. Real-Time Radiation Feedback System:

Implemented, tested & operated first bolometer radiation feedback at W7-X

- First controlled (P_{rad}), stable detachment with $f_{\text{rad}} \geq 85\%$
- Target heat load reduction factor ≥ 2
- Validated 3-5 channel LOS subsets for fast prediction ($\geq 80\%$ accuracy)

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3. Tomographic Reconstruction:

Benchmarked and tailored MFR algorithm with *RDA* weighting

- Extensive artificial emissivity profile testing & parameter variation
- Reconstruction quality $\geq 80\%$ for well-constrained phantoms
- Successful Application to experimental data & power balance calculations

OUTLOOK

FUTURE AND RECOMMENDATIONS

System Upgrades:

- Reduce feedback latency < 10 ms
- Implement machine learning for P_{pred} /LOS selection
- Per-experiment configuration (K_M , V_M)
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Tomography Extensions:

- Larger phantom benchmark set (> 50 cases) and parameter space
- Real-time algorithms (benchmark LUT?)
- Automated parameter optimization (grid search, ML)

COLLABORATIONS

Acknowledgement

This work has been carried out within the framework of the EUROfusion Consortium and has received funding from the Euratom research and training programme 2014-2018 under grant agreement No. 633053. The views and opinions expressed herein do not necessarily reflect those of the European Commission.



Collaborators:



Thank You

REFERENCES I